🎵 Music Features & Lyrics Analysis

Final Report

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Course: DSA210 – Introduction to Data Science

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1. Introduction

Music is a multifaceted form of expression shaped by both **acoustic characteristics** and **linguistic content**.
Digital music platforms such as Spotify describe songs through numerical audio features like energy, valence, tempo, and danceability. However, listeners often connect to music not only through sound but also through **lyrics**, which convey emotional tone, repetition, and narrative structure.

Despite this, most large-scale music analysis and recommendation systems rely primarily on surface-level numerical features. This raises an important question: **how much can actually be inferred from commonly used musical and lyrical representations?**

This project investigates the relationship between **audio features**, **lyrical features**, and song-level attributes such as **popularity**, **emotional valence**, and **energy**.
Rather than focusing on achieving high predictive accuracy, the project aims to explore the **strengths and limitations** of these representations and to interpret cases where they fail.

The final dataset consists of approximately **1,700 songs**, obtained by merging two public Spotify-related datasets.

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2. Data Collection and Pre-processing

2.1 Data Sources

Two publicly available Kaggle datasets were used:

- **Spotify Tracks Dataset**, containing numerical audio features such as energy, valence, tempo, loudness, and danceability.

- **Audio Features and Lyrics Dataset**, containing full song lyrics along with track and artist metadata.

2.2 Pre-processing Steps

The following steps were applied:

1. Merge the two datasets using shared track-level identifiers.
2. Remove instrumental tracks and songs with insufficient lyrical content.
3. Handle missing values and standardize feature formats.
4. Filter very short songs that could bias lyrical statistics.

After cleaning and filtering, the final dataset included **~1,700 songs** with both audio and lyrical information.

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3. Feature Engineering

3.1 Musical Features

The analysis uses Spotify's standard numerical features:

- Energy
- Valence
- Tempo
- Loudness
- Danceability

These features aim to represent **musical intensity, mood, and rhythm**.

3.2 Lyrical Features

Lyrics were processed using basic NLP techniques. The extracted features include:

- Total word count
- Unique word count
- Lexical diversity
- Average word length
- Repetition-related indicators

These features capture **structural simplicity and repetition**, rather than deep semantic meaning.

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4. Methods

4.1 Research Questions

The analysis focuses on the following questions:

1. Can song popularity be predicted using musical features?
2. Can song popularity be predicted using lyrical features?
3. Are simpler and more repetitive lyrics associated with higher popularity?
4. Can emotional valence be inferred from lyrical features?
5. Can song energy (high vs. low) be predicted using lyrics alone?

4.2 Analytical Approach

- **Exploratory Data Analysis (EDA)** to understand feature distributions and relationships.
- **Statistical analysis** using correlations and group comparisons.
- **Machine Learning models**:
  - Linear Regression
  - Logistic Regression
  - Random Forest

4.3 Evaluation Metrics

- **Regression tasks:** R^2 and RMSE
- **Classification tasks:** Accuracy and confusion matrices

All models were compared against simple **baseline predictors** (mean or majority class).

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5. Results

5.1 Popularity Prediction

Neither musical nor lyrical features were able to reliably predict song popularity.

- Musical features performed only slightly better than baseline.
- Lyrical features alone performed similarly or worse.
- Combining both feature sets did not yield meaningful improvement.

This indicates that popularity is influenced by factors beyond the available data, such as marketing strategies, social trends, and platform algorithms.

5.2 Lyrical Simplicity and Popularity

A weak but consistent relationship was observed between **lyrical simplicity** and popularity:

- Songs with simpler and more repetitive lyrics tend to be slightly more popular.

However, the effect size is small, suggesting that lyrical simplicity alone is insufficient to explain popularity.

5.3 Emotional Valence Prediction

Lyrical features showed **moderate success** in predicting emotional valence:

- Models performed above baseline but with limited accuracy.
- This suggests that lyrics encode emotional tone to some extent, even without semantic embeddings.

5.4 Energy Prediction from Lyrics

Attempts to classify songs as **high-energy vs. low-energy** using only lyrical features failed to outperform baseline models.

This highlights that song energy is primarily determined by **musical and production characteristics**, not lyrical structure.

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6. Discussion

Several important insights emerge from the analysis:

- **Null results are informative:** Poor predictive performance reveals the limitations of surface-level features.
- **Lyrics convey mood, not intensity:** Emotional valence can be partially inferred, while energy cannot.
- **Popularity is multi-factorial:** Success depends heavily on external and contextual factors not captured in the dataset.
- **Model limitations:** Low or negative R^2 values indicate that only a small portion of variance is explained by the chosen features.

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7. Limitations and Future Work

Limitations

- Lyrical features are surface-level and do not capture semantic meaning.
- Popularity reflects platform-specific dynamics rather than intrinsic musical quality.
- Temporal trends and cultural context are not considered.

Future Work

- Use contextual or transformer-based lyric embeddings.
- Analyse popularity over time.
- Incorporate listener behaviour or playlist placement data.
- Explore cross-genre or cross-cultural differences.

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8. Reproducibility and AI Disclosure

All analyses were conducted in Python using Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, and NLTK.
The repository contains fully reproducible notebooks that can be run top-to-bottom.

AI-assistance disclosure:
AI tools were used to assist with code organisation, explanation of methods, and language refinement. All analysis design, implementation, and interpretation decisions were made by me.